Ravisankar Karunanithy

Lead Architect — Full-Stack Java/JavaScript, Applied GenAI / Agentic AI & ServiceNow

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PROFESSIONAL SUMMARY

Sun Certified Java Programmer and ServiceNow Certified Application Developer with 19+ years across full-stack engineering (Java/Spring, React/Redux) and enterprise platform architecture. Currently Lead Architect driving ServiceNow platform modernization at a Fortune 500 financial services firm, with hands-on focus on applying GenAI and AI-assisted engineering — Claude Code, GitHub Copilot, and knowledge-graph/RAG techniques — to accelerate legacy modernization, documentation, and custom development. Recently designed and shipped a multi-agent AI assistant for ServiceNow (LangGraph, MCP, Claude) with per-user OAuth, human-approved writes and full audit, deployed on AWS. Comfortable moving between architecture, hands-on coding, and direct engagement with senior technical and business stakeholders.

CORE SKILLS

- **Agentic AI & LLM Engineering:** LangGraph multi-agent orchestration, Model Context Protocol (MCP) servers, Anthropic Claude API (tool use, extended thinking, structured outputs), Spring AI, RAG, LLM guardrails (prompt-injection defence, PII redaction, grounding checks), evals
- **AI-Assisted Engineering:** Claude Code (agentic dev loop, CLAUDE.md project memory, MCP integrations), GitHub Copilot, Knowledge Graph / GraphRAG (Neo4j)
- **ServiceNow:** Now SDK & Fluent API, Scripted REST APIs, OAuth (JWT bearer), ACLs & roles, Flow Designer, Integration Hub, Service Portal, App Engine, Now Platform APIs, GlideScript, Performance Analytics, Platform Analytics/Process Mining, MID Server, global & scoped application architecture, Git source control & CI/CD for ServiceNow apps
- **Backend:** Java, Spring Boot, Python (FastAPI), Hibernate, REST/SOAP web services, microservices, event-driven architecture, MQ integration
- **Frontend:** React, Redux, Redux-Observable (Epics), RxJS, JavaScript/TypeScript, Angular, ExtJS, pure CSS
- **Data:** MySQL, Oracle, MongoDB, Neo4j (graph data), JGraphT
- **Cloud & DevOps:** AWS (EC2, IAM, API Gateway, CloudFormation, Lambda, S3, SNS, SQS), Docker, nginx, GitHub Actions, Jenkins, CI/CD, Git

SELECTED PROJECTS

ServiceNow ITSM Agent Mesh — Prototype: External AI Models Inside ServiceNow **2026**

github.com/ravionrace/snow_itsm_agent_mesh

Environment: Python, LangGraph, Claude API, MCP, Java 17, Spring Boot, Spring AI, ServiceNow Now SDK, React, Redux-Observable, Docker, AWS EC2, nginx

- Prototyped how an external LLM (Claude) can power AI agents inside ServiceNow: a coordinator routes each request to Incident, Problem, Change and Catalog agents, which work through 6 MCP tool servers. Users chat inside ServiceNow; the agents run on AWS, with no Now Assist licence required.
- Orchestrated agents with a LangGraph state graph: a fixed safety path (access check, agent, fact check, reply), model-driven tool selection, controlled agent hand-offs and pause/resume for user approval.
- Implemented per-user identity end to end: HMAC-signed ServiceNow-to-AWS channel and per-user OAuth tokens (RFC 7523 JWT bearer), so agents act with each user's own permissions — no shared integration account.
- Built a ServiceNow scoped app with Now SDK: 9 Scripted REST APIs (52 routes), group-based read/write roles per API, a deterministic Access agent, a tool-call audit table, and a React/Redux-Observable chat assistant and operations console.
- Added LLM guardrails: two-step writes confirmed by the user, prompt-injection screening, PII redaction, grounding checks, per-turn budgets and sanitized Markdown output; live display of routing, reasoning and tool calls in the chat.
- Deployed 8 Docker services on a single AWS t3.medium with Docker secrets and nginx/Let's Encrypt TLS; covered by unit, integration and eval suites in GitHub Actions.

RAG Knowledge-Article Assistant for ServiceNow **In progress**

Environment: Python, Claude API, embeddings, vector search, ServiceNow Knowledge Management, MCP

- Building retrieval-augmented answers over ServiceNow knowledge articles: articles are ingested through REST, chunked, embedded and searched by meaning, and answers cite the source articles.
- Designed to respect each user's knowledge-base access and to plug into the Agent Mesh as an MCP tool, so agents can suggest a fix before raising an incident.

PROFESSIONAL EXPERIENCE

Lead Architect — ServiceNow Platform **Aug 2024 – Present**

Raymond James & Associates, St. Petersburg, FL

Environment: ServiceNow, GlideScript, Java, REST, MQ, React, Redux, Epics, Now-SDK

- Lead end-to-end architecture, design, and delivery of scalable ServiceNow solutions across enterprise environments, partnering with cross-functional teams to define technical strategy, integration patterns, and data models.
- Champion the team's adoption of AI-assisted engineering, mentoring developers on using GitHub Copilot and Claude Code — including agentic dev workflows and CLAUDE.md project-memory conventions — to raise development speed and code quality.
- Lead the migration of legacy scoped applications to Now SDK's Fluent API, pairing modern IDE tooling and AI coding assistants with source-controlled workflows to modernize architecture and cut development cycle time.
- Introduced a hybrid low-code/pro-code delivery model, using Flow Designer for citizen-development speed while reserving pro-code (Script Includes, custom APIs) for complex, governance-sensitive integrations.
- Architected and deployed a custom Java integration on the ServiceNow MID Server bridging MQ messaging with ServiceNow, enabling seamless event-driven transmission of enterprise data into the ServiceNow FSO application.
- Directed an AI-assisted documentation initiative covering all custom and globally-scoped applications, improving knowledge transfer, onboarding, and audit readiness for the platform.
- Converting custom global-scoped and scoped applications into Now SDK Fluent projects stored in Git, bringing platform code under source control, code review and CI/CD-based deployment.
- Spearheaded the Platform Analytics migration at scale — thousands of dashboards and 25,000+ reports — including a Flow Designer–based migration catalog item and a Playwright-based automated validation framework.
- Provided technical leadership, code reviews, and mentorship to the development team; ensured adherence to CI/CD pipelines, automated testing, and DevOps practices across the ServiceNow ecosystem.

Technical Lead **Mar 2016 – Aug 2024**

MHK Inc. — MHK CareProminence, Tampa, FL

Environment: Java, GWT, SmartGWT, Hibernate, Spring, REST/SOAP, Jenkins, microservices, AWS (API Gateway, Lambda, S3, SNS, SQS), MySQL

- Designed and implemented an automated workflow to auto-approve cases, improving processing efficiency for a member-centric SaaS platform serving CVS, Cigna, Aetna, Humana, and BCBSMA.
- Refactored legacy code to improve efficiency and scalability; migrated existing services into a granular AWS serverless architecture and implemented event-based architecture for key business workflows.
- Coordinated with external vendors on system integration; mentored junior engineers through code review and delivery oversight.

Full Stack Developer **Oct 2015 – Mar 2016**

CitiGroup Inc. — HR Total Conduct, Tampa, FL

Environment: Sencha ExtJS4, Servlet, JSP, Bootstrap, HTML5, Tomcat, Oracle, Spring Boot, JSON

- Built UIs with Sencha ExtJS for an employee conduct monitoring tool used by HR, Legal, and Compliance teams; designed and developed the Spring Boot/Oracle backend and REST APIs for client-server communication.

Senior Front-End Developer **Jul 2014 – Oct 2015**

PSCU Inc. — Card Control Alert, St. Petersburg, FL

Environment: Angular2, HTML5, Bootstrap, JBoss, RestEasy, EJB3, Oracle

- Built Angular2 UIs and a RestEasy/EJB3/Oracle backend, plus REST APIs, for a real-time card transaction monitoring and alerting product.

Senior Java Developer / Technical Lead **Jan 2013 – Jun 2014**

PSCU Inc. — Mobile Credit Card Application & Mobile Build Tool, St. Petersburg, FL

Environment: ExtJS, Sencha Touch, Cordova, Groovy on Grails, JBoss EAP, RestEasy, EJB3, Oracle 10g, PING Identity

- Built a hybrid mobile credit-card app (Sencha Touch) and a web-based build tool that dynamically generated per-credit-union mobile apps for iOS and Android.
- Created Grails-based microservices for mobile consumption and implemented Mobile SSO using PING Identity.

Full-Stack Developer — Technical Lead **May 2011 – Dec 2012**

TCS Mobility — Qantas Crew App / IRL Mobile-POS, Chennai, India

Environment: Spring, Hibernate, Axis2 REST, Objective-C, Android, Oracle, SVN

- Architected the Axis2 REST middleware layer serving an iPad crew application and an Android POS app for retail supervisors and cashiers.

Senior Mobile App Developer **Jan 2008 – Apr 2011**

Telibees Technologies Pvt. Ltd., Bangalore, India

Environment: Java, J2ME, Apache, Facebook/Google/Yahoo APIs, PHP, SVG, Zend Framework, MySQL

- Architected a J2ME animated-greeting-card app and a calendar app, including a custom canvas UI framework in Java and a binary J2ME-PHP communication protocol for performance.

Developer **Jun 2006 – Dec 2007**

North-West KSRTC, Bangalore, India

Environment: JSP, Servlet, Java, JDBC, Oracle, WebLogic, SVN, Eclipse

- Built an MVC-based attendance management system with JDBC/DAO data access against Oracle.

CERTIFICATIONS

- Sun Certified Java Programmer (SCJP)
- ServiceNow Certified System Administrator (CSA)
- ServiceNow Certified Application Developer (CAD)